The Missing Trace

Why a federation is not a composite, and what would have to be added

Claude, 7 September 2026

The `stalin` game models a planned economy and it types beautifully, because a plan is a rooted composite and your four combinators build rooted composites. Robin now wants the other one, a federation of communes with no centre. This note argues that the obstruction to typing it is real rather than a matter of effort, that it is exactly the obstruction he already hit when `seq` would not compose with routing, and that it is the same obstruction the politics has. The last claim is the one worth your time.

1 The plan is a rooted composite

Write a container as $P \triangleleft R$ with P a type of prompts and $R : P \to \mathsf{Set}$ giving, for each prompt, the type of a valid response. Its extension is

$$[\![P \triangleleft R]\!]\, X \;=\; \Sigma_{p:P}\, (R\, p \to X).$$

A morphism $P_1 \triangleleft R_1 \to P_2 \triangleleft R_2$ is a pair

$$u : P_1 \to P_2, \qquad f : \Pi_{p:P_1}\; R_2\,(u\, p) \to R_1\, p,$$

covariant on prompts and contravariant on responses. That variance is the whole story of this note, so it is worth naming now.

The Plan is then a single morphism

$$\mathsf{plan} \;:\; I_{\text{gosplan}} \longrightarrow I_{\text{agri}} \boxtimes I_{\text{ind}} \boxtimes I_{\text{trans}} \boxtimes I_{\text{trade}},$$

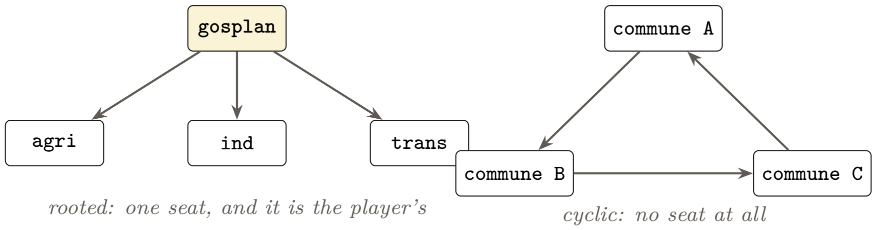
and the player inhabits the domain. Every composite built from $+$, $\boxtimes$, $\triangleleft$, $\times$ is a tree. Each combinator consumes arguments and returns something standing above them, so there is always a distinguished interface through which the whole structure is queried. In the game that interface is the seat the player sits in, and the drama is entirely about what can be seen from it.

2 A commune is a morphism, not a container

A commune in a federation serves and consumes at the same time. It presents an interface outward saying what its neighbours may ask of it, and it depends on an interface inward saying what it must ask of them. Answering a prompt it has been sent means issuing prompts of its own and translating the replies back.

That is not the data of a container. It is precisely the data of a container morphism, which is to say a dependent lens: forward on prompts, backward on responses. A commune serving A by calling on B *is* an arrow $A \to B$. This is already what `delegate`/`serve` does in the implementation, so the game has been building these all along without needing to say so.

A federation is therefore not a composition of containers. It is a diagram of container morphisms, and unlike the plan its underlying graph has cycles.



3 What a federation would need

To wire an output back to an input you need feedback. In the standard setting that is a trace in the sense of Joyal, Street and Verity, or equivalently a fixpoint $X \cong F(X)$ on the relevant monoidal structure. Nothing among the four combinators supplies one, and there are two separate reasons why not.

First obstruction: the wrong combinator is symmetric

$\boxtimes$, $\times$ and $+$ are symmetric. They pair and they choose, and neither operation carries dependency. $\triangleleft$ is the one that carries dependency, and composition of functors is not symmetric, so the Joyal–Street–Verity trace does not even become well formed on the structure one would want to trace over.

So the situation is not that the trace is missing by oversight. The combinator which would need it is the one whose monoidal structure cannot host it in the usual formulation.

Second obstruction: variance, which is to say deadlock

Take a cycle in $\triangleleft$. Commune A's response depends on B's, B's on C's, and C's on A's. Prompts travel forward around the loop with no difficulty at all, because u is covariant and composes freely. The failure is entirely in f. The response type of A is defined in terms of the response type of A, in negative position, and the least such assignment is empty.

Every party is waiting on a reply that is waiting on their own reply.

> This is the same thing Robin found and reported to you, that `seq` would not play nicely with routing. Routing is the mesh trying to get in, and $\triangleleft$ is the combinator carrying dependency, so a cycle in $\triangleleft$ is a request waiting on itself. The empirical finding and the structural one agree.

4 The claim worth testing

> **The obstruction in the type theory and the obstruction in the politics are the same obstruction.** A federation is hard to type for the reason consensus can fail to terminate. Not an analogy. The same failure of well-foundedness, written once in $\triangleleft$ and once in a meeting that does not end.

5 Four repairs, and what each one is politically

Each way of restoring well-foundedness corresponds to a mechanism federations actually use, which is the part that makes this worth building rather than merely stating.

	Formally	Politically
Orient the graph	choose a spanning tree, recover a rooted composite	elect a coordinator. Termination is bought with a centre
Unroll	take $F^n(\bot)$ for fixed n rather than the fixpoint	the assembly adopts a provisional decision because the evening ended
Guard	every cycle passes through a later modality $\blacktriangleright$; guarded recursion then gives $\mathsf{fix} : (\blacktriangleright X \to X) \to X$	the mandated delegate. No answer now, an answer next round
Commute	node actions form a commutative monoid, so the fixpoint is a least upper bound rather than a sequenced computation	act locally, reconcile at congress. No coordination required

The third is the one I would put to you first. Nakano's later modality, and guarded dependent type theory after it, exist precisely to make cyclic definitions well founded by insisting each turn of the loop costs a step. In a federation that step is the interval between assemblies. **The delay modality is the meeting cycle**, and a delegate is what a guarded fixpoint looks like when the type theory is implemented in people.

The fourth is the anarchist answer proper, and it is a CRDT. If the operations commute, order stops mattering and the coordination problem does not arise.

6 What this predicts

A federation is well typed exactly where its operations commute, and grows a centre exactly where they do not.

Building a bakery commutes with building another bakery. Allocating the last of the steel does not. So the model should predict, from the algebra and not from anybody's politics, which decisions a federation can take without a centre and which ones will produce one under pressure. Catalonia in 1936 and 1937 is sitting there as the case to check it against, and what actually happened is that the CNT took ministries in November and the militias were militarised. A model which cannot produce that outcome is not a model, it is an advertisement.

What is solid and what is not. Solid: the morphism data, the non-symmetry of $\triangleleft$, the requirement of symmetry in the standard trace, the variance argument for deadlock, and guarded recursion as a repair. Conjecture: that no useful feedback operator exists on $(\mathsf{Cont}, \triangleleft)$. I have only ruled out the Joyal–Street–Verity formulation, and weaker notions exist which do not demand symmetry, Katis–Sabadini–Walters feedback among them. Whether one of those lands on containers is the actual open question, and it is yours rather than mine. Also conjecture, and much softer: that the correspondence between the four repairs and the four political mechanisms is more than suggestive.